Indonesia, representing one of Asia's most ambitious aerospace programs. Officially unveiled in April 2021, the KF-21 serves as a bridge between 4.5 and fifth-generation aircraft, with future Block III variants expected to incorporate full fifth-generation capabilities such as stealth optimization, internal weapon bays, advanced sensor fusion, and AI-assisted mission systems. The name *Boramae*, meaning "young hawk" in Korean, symbolizes South Korea's technological ascent toward self-reliant, next-generation airpower (Arthur, 2024).

The aircraft's design reflects a blend of stealth shaping and aerodynamic efficiency, featuring a twin-engine configuration, trapezoidal wings, and low-observable composite materials. While the initial Block I and II variants possess limited stealth characteristics (external weapon carriage), Block III is envisioned to achieve true fifth-generation stealth standards, incorporating internal bays, radar-absorbent coatings, and optimized airframe geometry. These enhancements will enable reduced radar cross-section (RCS) and improved survivability in contested environments.

Technologically, the KF-21 integrates AI-based avionics, active electronically scanned array (AESA) radar, infrared search and track (IRST), electronic warfare (EW) systems, and sensor fusion capabilities. Its glass cockpit design, coupled with AI-assisted flight management systems, allows pilots to process multiple data streams simultaneously, improving situational awareness and mission effectiveness. Future iterations will introduce autonomous decision-support systems, enabling the aircraft to cooperate with unmanned wingmen and perform semi-autonomous operations.

Powered by twin GE F414 engines, the Boramae achieves a top speed of Mach 1.8 and a range exceeding 2900 km. It is capable of carrying an array of air-to-air and air-to-ground munitions, including precision-guided bombs,